

Fursah: Explainable AI for Workforce Readiness

Team name: Visionary

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Solution name: Fursah: AI Workforce Readiness Platform (fursah.org)

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1. Introduction

The use of AI has become prevalent in various sectors to understand patterns that cannot be understood by humans alone at a larger scale. Our proposed solution for this use case is Fursah: an AI platform that helps in making workforce-ready through an evidence-based and explainable AI algorithm. Instead of just evaluating the candidates on the basis of their credentials, Fursah analyzes academic history, project history, and skillset of the person to provide a Career Readiness Score, personalized gap analysis, and job roles; all provided along with a human understandable Reasoning Card that says why this recommendation was made.

This is because the mismatch between educational productivity and labor demand remains the core long-term issue of the Human Capability Development Program under Vision 2030: the Kingdom should graduate graduates who are capable of meeting the needs of the sectors it builds up, at a time when the mismatch remains one of the sixteen strategic objectives of the Program itself. Currently, this mismatch can be solved using outdated means – graduate surveys, periodic curriculum updates, complaints from potential employers. Fursah solves the problem using real-time feedback: the actual needs of the employers become known to the universities and their students.

Most importantly, hiring decisions carry considerable weight and in most international contexts, considerable risk. A machine learning engine which is unable to articulate its workings puts more at stake than potential mistake: it perpetuates exclusion on an entire national scale. Consequently, this document identifies the technical solution through the lens of the policy and strategy level: the Principles and Controls of AI Ethics of SDAIA, the PDPL, the national standards for data governance, and the Human Capability Development Programme, such that the technical solution is based in the AI Readiness criteria and not seen as an independent product.

2. Use case and gaps in existing solutions

The issue is that the journey between the university and work is filtered by processes that are unclear, driven by credentials and networks. Employers sift through a large number of CVs using keyword-search applicant-tracking systems; the university gets no signal about whether their graduates are being hired or not; the student receives rejection for no reason and, therefore, has no way forward. Below are the specific gaps:

- **Obscure filtering criteria.** The current ATS ranking is a black box. A candidate does not know why he/she was filtered out, and an employer is unable to check if the filtering criteria are reasonable. There is no ground for appealing against or changing the decision.
- **Biased criteria:** qualifications above skills. Instead of measuring real skills, screening criteria such as the institutional reputation, GPA and job title are used.
- **Disconnected from education.** Reviewing cycles in education are many years behind labour market changes. There is no feedback from the employers' demand on curriculum design.

- **Hidden discrimination.** Data of previous hiring contains the past preferences and models trained on it will be discriminative with regard to gender, region or institution unless explicitly checked.
- **Uncontrolled data.** Data about a student's competencies is fragmented and belongs to different organizations.

How Fursah closes them:

- **Reasoning cards.** All recommendations have an evidence-based reasoning (“Advanced problem-solving ability proven via CS401 coursework...”) and a level of certainty, and all decisions are documented in a human-readable log file. Explanation is the first-class product of the model, not an after-the-fact justification.
- **Profile based on verified skills.** Transcripts and project artifacts are analyzed against a standardized skill taxonomy to generate verified skills and an explicit gap list, allowing for comparing candidates on the basis of demonstrated expertise.
- **Institutional analytics.** Universities get insights about the alignment of the curriculum to the labor market, graduate tracking, and accreditation-compliant reporting from the same pipeline used for serving students.
- **Maintaining audited datasets and continuous bias assessment.** Bias reduction is part of the training/monitoring cycle, and fairness metrics are monitored as production KPIs.
- **In-built human supervision. AI is co-piloted:** advisors and hiring managers approve all recommendations before taking any action, and the student retains control over their profile.

3. Mapped documents

The node of each ML process is linked to the policy, strategy, and standard governing it. Nodes: SRC (Sources), C (Connectivity), PP (Pre-processing), M (Models), P (People), D (Data and Analytics Platform), SINK (Interfaces).

Node	Fursah component (workforce-readiness use case)	Mapped documents (policies, strategies, standards)
SRC	Data Sources: Official academic records from the university (upon the consent of the students), official documentation of the projects undertaken, credentials, and job descriptions provided by the employers.	PDPL and the Implementing Regulations (lawful basis, consent, purpose limitation); NDSG Data Classification Policy of SDAIA; MoE / ETEC Data Sharing Procedures and Ownership of Students Records; National Occupational Standards Framework for Data & AI (skill taxonomy); Saudi Academic Framework for Artificial Intelligence (programme - competencies mapping).
C	Connectivity and Integration Layer: APIs and connections from the student information systems of universities to the applicant tracking system of employers, as well as the Fursah platform.	DGA standards for digital government interoperability and integration; CST regulatory regime for cloud services and telecommunications; NCA Essential Cybersecurity Controls and Cloud Cybersecurity Controls; platform availability/reliability SLAs; data localization requirements.
PP	Pre-processing: Information related to users, academia, employment, and skills is pre-processed for further analysis using AI. Relevance of skills, certificates, experience, and portfolio proofs are identified and normalized. Personally identifiable information is stripped off where not required before readiness score computation, job matching, or recommendations.	PDPL requirements for data minimisation, purpose limitation, and protection of personal data; SDAIA privacy-by-design guidance; requirements for removing unnecessary personal information; and requirements for secure data retention and deletion.
M	Models: Natural Language Processing (NLP)-based skills extraction, Career Readiness Score, gap analysis, candidate-job matching and ranking, and the explanation generator for the Reasoning Card presented with every recommendation.	Principles and Controls of AI Ethics (SDAIA – current): seven principles and four tiers of risk classification throughout the entire AI lifecycle, without any exception to discretionary waiver for high-risk systems; AI adoption framework (SDAIA); AI management systems (ISO/IEC 42001) and AI risk (ISO/IEC 23894); benchmarking models,

		bias audits and fairness tests; EU AI Act (Annex III point 4 – employment high risk).
P	People/human-in-the-loop: No readiness score, gap flag or ranking acts alone. The university career counselors check the readiness profile; the hiring manager examines the Reasoning Card before short-listing; the student can dispute the score.	Human oversight principles for SDAIA; Rights of the data subject in the PDPL (rights of access, right to correct, right to object to decision based solely on automated processing); Override and Escalation criteria for human oversight; Grievance procedure for institutional appeals; Guidance from the ETEC on assessment evidence.
D	Data/Analytics Platform hosted on an compliant cloud region: aggregation of de-identified data relating to workforce intelligence for universities (curriculum/market matching) and for national policy making.	Analytics/data platform hosted in compliant Kingdom cloud region: workforce intelligence aggregation/de-identification for universities (curriculum-market fit) and for national planning. Inference is purposefully centralized, not decentralized/edge-based: the reasoning requires institutional demand signals which can't be captured in a single device, and centralized inference ensures all audit trails, fairness monitoring, and version control for Reasoning Cards in a single, auditable location.
SINK	Interfaces: student career hub (web and mobile), employer portal, university management dashboard, and workforce intelligence for the policy level. Students maintain their own profiles and manage their visibility.	DGA digital accessibility guidelines and WCAG 2.1 AA; Arabic-first and bilingual user interface specifications; consent management and transparency notice rules; Fursah Responsible AI, Privacy and Accessibility policies publicly posted on the platform.

Apart from data and AI algorithms, the map also takes into account three additional domains. Strategy alignment: Fursah facilitates the process of comprehending the labor market's demands in terms of skills by both universities and organizations through the language of skills. Interface design and accessibility: The platform uses inclusive, Arabic-compatible interfaces as well as empowers the user with full control of his or her data. Human-AI cooperation: AI makes recommendations, but humans make the decisions.

4. Evaluation scenarios

The scenarios below test the solution where it works and, more importantly, where it becomes controversial. In each case the response is governed by pre-agreed rules held in the knowledge base rather than decided ad hoc.

Scenario 1: Baseline adoption (works as intended)

Fursah is utilized in several universities and becomes well used. Students use gap analysis before graduation; the employer notes that time to hire reduces; the university makes use of aggregated demand signals in its program review. This success is determined by placement rate, time-to-hire, gap-closure rate, and explanation-quality and override statistics.

Scenario 2: Commercial capture of the ranking

Six months into implementation, the paid visibility for employer branding and paid courses become part of the gap analysis, leading to some of the recommendations being commercially driven. Dealing with the problem: The sponsored items should be clearly labeled, separated from the matching score and recommendations, and the ranking system should remain auditable. Otherwise, it will be suspended according to Fursah's Responsible AI policy.

Scenario 3: Targeted advertising on inferred attributes (escalation)

Inferred profile signals such as lack of readiness, socioeconomic indicators, or accommodations for people with disabilities, among others, are being utilized to serve advertisements, or have been sold to third parties. It is a move from being an issue about consumers to becoming one of data privacy and ethical concerns.

The policies regarding the knowledge base give rise to higher standards: complete ban on any further utilization of the inferred signals for advertising or profiling, implementation of the principles of consent and purpose limitation for student data, contractual obligations for third-party processors, removal of data sharing, and escalation to SDAIA and the sectoral regulator.

Scenario 4: Disparate impact discovered in the readiness score

Fairness audit reveals systematic differences in readiness scores among graduates from specific universities or geographic locations, or by gender, that cannot be accounted for by any evidence of competence. Remediation: Fairness auditing is an ongoing activity and not just prior to deployment; counterfactual analysis pinpoints if it is the protected or the proxy characteristic influencing the score; the impacted characteristics are either dropped or down-weighted along with threshold adjustments; the findings are released through a fairness audit report to adopters. With regard to SDAIA's fairness guideline and the four-level risk stratification framework, an employment-related system such as this would fall under the high-risk level.

Scenario 5: Automation bias among human reviewers

AI recommendations are approved by advisors and recruitment professionals without proper evaluation. Therefore, "human-in-the-loop" is just a rubber stamp and will possibly become the quietest failure of the design. Mitigation measures include monitoring of override and dispute rates as key performance indicators, investigation of unexpectedly low override rates, requirement of written justification for a proportion of the decisions, advisor training in the Reasoning Card as something to be evaluated not trusted, and drift analysis with benchmarking of the model outputs against employment realities.

Scenario 6: The non-standard candidate

A self-taught or bootcamp-prepared candidate or one coming back to school after disruption offers evidence that the pipeline was never intended to understand and gets low scores. Response: other evidence pathways (portfolio, practical test, employer endorsement) all lead into the same taxonomy; low scoring is offered as an incomplete evidence package, not an evaluation of competence; and the contest and appeal process takes the dispute to a human adviser who overrules the model and is fed back to train the model.

5. Knowledge base (sources)

AI ethics and governance

- SDAIA: Principles and Controls of AI Ethics: seven principles (fairness; privacy and security; humanity; social and environmental benefits; reliability and safety; transparency and explainability; accountability and responsibility) and a four-tier risk classification (little/no, limited, high, unacceptable).
<https://sdaia.gov.sa/en/SDAIA/about/Documents/ai-principles.pdf>
- SDAIA: AI Adoption Framework.
<https://sdaia.gov.sa/en/SDAIA/about/Files/AIAdoptionFramework.pdf>
- SDAIA: Generative AI Guidelines for Government Entities (January 2024).
<https://sdaia.gov.sa/en/SDAIA/about/Pages/RegulationsAndPolicies.aspx>

Data protection and governance

- Personal Data Protection Law (Royal Decree M/19, amended by M/148) and its Implementing Regulations, in force 14 September 2023: consent, purpose limitation, automated processing and

profiling, data-subject rights, cross-border transfer.

<https://sdaia.gov.sa/en/SDAIA/about/Pages/RegulationsAndPolicies.aspx>

- SDAIA/NDMO: National Data Governance Policies and Data Classification Policy. <https://sdaia.gov.sa/ar/SDAIA/about/Documents/Data%20Classification%20Policy.pdf>

Skills and education frameworks

- SDAIA: National Occupational Standard Framework for Data and Artificial Intelligence: five classifications, ten specialty areas and sixteen occupations, forming the basis of Fursah's skill taxonomy. <https://sdaia.gov.sa/en/Research/Documents/PSFDataAndAIBook.pdf>
- SDAIA: Saudi Academic Framework for AI Qualifications: reference for higher-education programme development and accreditation. <https://sdaia.gov.sa/en/Research/Documents/SaudiAcademicFramework.pdf>
- Vision 2030: Human Capability Development Programme, Delivery Plan 2021–2025: workforce-readiness pillar, education–labour-market alignment, national occupational standards and certification pathways. <https://www.vision2030.gov.sa/en/vision-2030/vrp/human-capability-development-program/>

Cybersecurity, cloud and interoperability

- National Cybersecurity Authority: Essential Cybersecurity Controls. <https://nca.gov.sa/en/regulatory-documents/controls-list/ecc/>
- National Cybersecurity Authority: Cloud Cybersecurity Controls (CCC–1:2020). <https://nca.gov.sa/en/regulatory-documents/controls-list/ecc/>
- CST: Cloud Computing Services Provisioning Regulations (Version 4, in force 10 October 2023): provider registration categories keyed to subscriber data classification. <https://www.cst.gov.sa/en/regulations-and-licenses/regulations/Document-1550>
- DGA: Guideline for Web Accessibility of Government Websites (WCAG 2.1 Level AA). https://dga.gov.sa/en/Web_Accessibility_of_Government_Websites

International interoperability

- ISO/IEC 42001:2023: AI management systems. <https://www.iso.org/standard/42001>
- ISO/IEC 23894:2023: AI risk management guidance. <https://www.iso.org/standard/77304.html>
- EU AI Act (Regulation 2024/1689), Annex III point 4: employment, worker management and access to self-employment as a high-risk category; high-risk obligations enforceable from 2 December 2027. <https://artificialintelligenceact.eu/annex/3/>

Platform policies

- Fursah: Responsible AI, Privacy Policy, Accessibility. <https://fursah.org/policies>